# Kellen Downey

Business analytics · Defense-market research · Quantitative research

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| **USC AI Case Competition** Semifinal round | **General Atomics** Strategic Development | **WPSA 2025** Research presenter | **3.7** Undergrad GPA |

## EDUCATION

**University of Southern California, Marshall School of Business** — M.S. Business Analytics — Expected Dec 2027

- Coursework: Python, SQL, multivariable regression, A/B testing
- Activities: Business Analytics Student Association · USC Symphony Orchestra (cello)

**Chapman University** — B.A. Economics & B.A. Political Science (Political Economy emphasis) — May 2026

- Certificate of Applied Statistical Analysis, A. Gary Anderson Center for Economic Research
- Presidential Scholarship · Order of Omega · Pi Sigma Alpha · Dean's List 2022, 2024
- Delta Tau Delta: elected Honor Board representative (judicial/conduct); organized the chapter's most successful philanthropic event

## PROJECTS

**Podcast Opportunity Explorer** — Team M7 (5 members), DSO 576, USC Marshall — Fall 2026

- Developed an individual product vision and contributed to a team screening tool for a podcast-acquisition manager (Python, pandas, SQLite, Streamlit) built on the Podcast Index (~4.7M feeds; 6,000-row development sample).
- Audited the blank-title diagnostic with an independent Python script (16 of 6,000 titles, 0.27%); proposed two change requests.
- Defined two property-based tests (4 cases across pandas and SQLite) for the instructor-supplied build: shuffled rows never change the shortlist, and stricter filters never add feeds (suite grew from 45 to 49 passing tests).

**Trojan Deadline Hub** — USC Marshall Global AI Case Competition — Fall 2026

- Co-designed an AI academic-planning concept with a six-person team (one of 44 semifinal-round teams) that consolidates syllabus deadlines, exams, and grade weights into one calendar.
- Design requires student verification of every AI-extracted date, flags missing or TBD dates instead of guessing, and adds a workload estimator that suggests start dates and splits large assignments into study sessions.

**Shared-Housing Revenue Opportunities** — two-person team project — Fall 2025

- Co-authored a property-analytics study (465 properties, 99 engineered features): compared regression and classification models and translated results into a Top-20 shortlist and a proposed 90-day pricing pilot.

## EXPERIENCE

**Strategic Development Intern** — General Atomics Aeronautical Systems, San Diego, CA — Jun – Aug 2025

- Conducted defense-market research and analysis that supported leadership's decision to pursue a new unmanned-aircraft market.
- Analyzed defense-program budgets, ran regression analyses, built Power BI dashboards, and drafted contract-proposal materials.

**Graduate Assistant, Graduate & International Programs** — USC Gould School of Law — Fall 2026 – Present

- Conduct first-pass reviews of graduate applications against admissions criteria; coordinate applicant communications and workflow.

**Office Assistant, Office of Admission** — Chapman University — Sep 2024 – May 2026

- Processed and audited application materials; selected as 2026 Admission Student of the Year by the admission counseling staff.

**Legal Intern** — Children's Law Center of California, Monterey Park, CA — Aug – Dec 2024

- Synthesized approximately 50 case files per day, briefed staff attorneys, and supported client intake interviews.

**Earlier:** Sweetgreen, 2024 (corporate liaison; helped pilot Infinite Kitchen automation) · Chick-fil-A Shift Lead, 2019–2023

## RESEARCH EXPERIENCE

**Assassination Attempt: Media Exposure & Public Opinion** — Chapman, POSC 399 (Dr. Lewis Luartz) — 2024 – 2025

- Co-authored a multi-semester study (manuscript submitted for publication) of how social-media news exposure was associated with candidate evaluations before and after the July 2024 assassination attempt, using two ANES surveys; contributed statistical analysis.
- Team's R workflow combined multiple imputation, propensity-score matching, ordinal models, and marginal effects; presented early findings at the Western Political Science Association 2025 conference.

**Loneliness & Support for Political Violence** — independent capstone, Chapman, POSC 410 (Dr. Ann Gordon) — Fall 2025

- Analyzed Chapman Survey of American Fears data: built a loneliness index, fit logistic models with controls and age interactions, and found no evidence that loneliness increased support for political violence; poster in Chapman Digital Commons.

## SKILLS

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| **Programming & data** | Python (pandas, pytest), SQL and SQLite (DBeaver), R, Stata, Streamlit, Git/GitHub |
| **Analytics** | Multivariable regression (linear, logistic) · market segmentation · predictive modeling · A/B-test interpretation |
| **Tools** | Power BI, Excel (dashboards), Bloomberg Terminal, generative AI for coding and research |
| **Interpersonal** | Cross-cultural collaboration · teamwork · leadership · empathy · emotional intelligence |